

Nonlinear optimization

Numerical tasks

1. (2 points)

Create a synthetic dataset for sparse linear regression:

- Generate $k \geq 100$ normally distributed d -dimensional random variables $X_1, X_2, \dots, X_k$, where $d \geq 10k$,
- Select around 10 – 20% of these variables and define some linear combination to generate a target variable:

$$Y = \beta_1 X_{i_1} + \dots + \beta_n X_{i_n},$$

- Add some noise to the target.

Use this dataset to fit a linear regression model with L_1 penalty.

Compare three different algorithms:

- subgradient descent,
- proximal gradient descent (ISTA),
- proximal gradient descent with momentum (FISTA).

Plot the values of MSE at each step.

Note. For more details about ISTA and FISTA, you can look at this presentation:

<https://www.stat.cmu.edu/~ryantibs/convexopt/lectures/prox-grad.pdf>

2. (1 point)

The *Alternating projections* algorithm is a simple way to find points in the intersection of two convex sets. It works by repeatedly projecting the current point on set $A, B, A, B, \dots$.

You can find more details here: https://web.stanford.edu/class/ee392o/alt_proj.pdf.

Consider two disks on a plane, one centered at point $(0, 0)$ with radius $r_1 = 1$ and one centered at point $(3, 0)$ with radius $r_2 = 2.05$. Choose a starting point (not belonging to either disk) and use the *Alternating projections* algorithm to find some point in their intersection. Visualize the results for a few different starting points.